Question 3: Time Series Forecasting - Cumulative GitHub Stars

3.1 & 3.2.1 Data Preparation

Dataset Selection: We analyzed the dataset and selected two repositories with sufficient historical data for modeling:

1. `pallets/flask` (Long-term stable growth)
2. `facebook/react` (High-growth popular library)

Preprocessing Pipeline: To ensure valid forecasting, we implemented a rigorous cleaning pipeline (Version 2.0):

1. **Artifact Removal:** Detected and truncated the artificial "4000 star ceiling" found in the raw dataset.
2. **Stationarity Transform:** Converted the non-stationary "Cumulative Stars" into "Daily Incremental Stars" ($y'_t = y_t - y_{t-1}$).
3. **Outlier Trimming:** Removed the first 30 days of "Viral Launch" phase where growth was volatile (+500/day) compared to the stable state (+20/day).
4. **Smoothing:** Applied a 3-day rolling mean to reduce high-frequency daily noise.

3.2.2 Model Architectures

We compared Classical and Deep Learning approaches. All models were trained on the incremental data to ensure stability.

Model	Architecture / Specs	Loss Function
Classical	**ARMA(5,0,5)** with Robust Fallback. *Strategy:* Fits high-order autoregression on incremental changes.	Likelihood / AIC
RNN	**SmallRNN**: 1 Layer, Hidden Size=32. *Input:* Sliding window ($k = 14$). *Output:* Linear decoder.	MSE (Mean Squared Error)
CNN	**1D CNN**: 1 Layer, 32 Filters, Kernel=3. *Pooling:* MaxPool1d(2). *Flatten:* Linear decoder.	MSE (Mean Squared Error)

3.2.3 Quantitative Results & Evaluation

We evaluated models using **Root Mean Squared Error (RMSE)** on the cumulative reconstruction.

Summary of RMSE Results (Lower is Better):

Repository	ARMA (Classical)	RNN (Deep Learning)	CNN (Deep Learning)	**Winner**
Flask	167.43	**32.86**	44.75	**RNN**
React	323.18	**55.83**	62.07	**RNN**

Findings:

1. **Deep Learning Superiority:** Contrary to initial expectations, the **SmallRNN** model achieved the best performance, with an RMSE roughly **5x lower** than the Classical ARMA model.
2. **Non-Linearity:** This suggests that even after smoothing, the daily growth patterns of these repositories contain non-linear dependencies that the ARMA model (which assumes a linear relationship between past and future) could not capture effectively.
3. **RNN vs. CNN:** The RNN consistently outperformed the CNN (32.86 vs 44.75 for Flask). This aligns with the theory that RNNs are architecturally better suited for sequential time-series data where the "memory" of the previous state is crucial, whereas CNNs focus on local patterns.